Quantized Multimodal LLMs Say Less, Not More: Locating and Repairing Text-Side Grounding Loss

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https://github.com/suryamanoj4/MLLM-quantization-research

Abstract

Multimodal LLMs are increasingly compressed to 3 bits per weight and below, and the prevailing expectation is that quantization noise weakens the visual evidence and makes models hallucinate. We test this on LLaVA-1.5 and Qwen2.5-VL with five standard quantizers and find the opposite: quantized models do not invent objects, they omit them. At 3 bits, captions name 6–7 points fewer of the objects in the image, small objects first, while hallucination metrics hold steady or improve and POPE does not move. Causal interventions locate the loss on the language side: quantizing weights only at image positions costs nothing, only at text positions everything, and restoring full-precision activations shows the loss building up across depth at text positions and peaking in the last MLPs. A closed-form, low-rank correction applied only at text positions removes the significant loss on both models; applied at every position, the same correction recovers less or breaks yes/no answering.

1 Introduction

Multimodal LLMs (MLLMs) such as LLaVA (Liu et al., 2023, 2024a) and Qwen2.5-VL (Bai et al., 2025) describe images and answer questions about them, and serving them means fitting a 7B-parameter decoder, 14 GB in 16-bit floating point (FP16), into the memory of a GPU, a laptop or a phone. Weight quantization is the standard answer, and its frontier is moving below 4 bits. Four-bit weights are now close to lossless for LLMs (Dettmers and Zettlemoyer, 2023; Jin et al., 2024), and research has turned to 3 bits and fewer: 3-bit models that scale better than 4-bit ones (Tseng et al., 2024), Pareto-optimal compression to 2–3 bits per parameter (Egiazarian et al., 2024; Kim et al., 2024), and 2-bit and ternary models on the size–accuracy frontier (Liu et al., 2025b; Ma et al., 2024). MLLM quantizers follow, with 3-bit weights (Li et al., 2025), sub-4-bit layer-wise budgets (Bhatnagar et al., 2026) and 2-bit on-device models (Guo et al., 2026). This is also where quantization stops being harmless: accuracy losses become moderate to high for 1–10B models (Dettmers et al., 2024), LLM trustworthiness drops significantly at 3 bits while 4-bit models retain it (Hong et al., 2024), and MLLMs collapse at 2 bits (Bhatnagar et al., 2026). Yet MLLM quantizers are judged by benchmark accuracy (Wang et al., 2024; Li et al., 2025; Yu et al., 2025), and methods against object hallucination are built and tested at full precision (Leng et al., 2024; Huang et al., 2024; Liu et al., 2025a). What low-bit quantization does to *which objects a model reports seeing*, and where in the network that change arises, is not known.

The natural expectation is that it makes them hallucinate. The literature on MLLM hallucination traces invented objects to one cause: language priors overriding weak visual evidence. Models over-rely on language priors at the expense of the visual context (Guan et al., 2024; Zhu et al., 2024); parametric knowledge in the language model overrides the visual content (Bai et al., 2024); models over-trust a few summary tokens and neglect the image tokens (Huang et al., 2024), or let the dominant language model drown out the vision encoder (Liu et al., 2024b). Crucially, *degrading* the visual input makes this worse: visual uncertainty pushes models to overlook visual evidence and exploit language priors, producing more hallucination, which is the very mechanism visual contrastive decoding builds on (Leng et al., 2024). Quantization degrades representations, and MLLM quantization work singles out the image tokens, whose activations span a much wider range than those of text tokens (Yu et al., 2025). In LLMs, quantization lowers truthfulness (Li et al., 2024) and 3-bit weights significantly reduce trustworthiness (Hong et al., 2024), and benchmarks of compressed MLLMs warn that simple quantization schemes may intro-

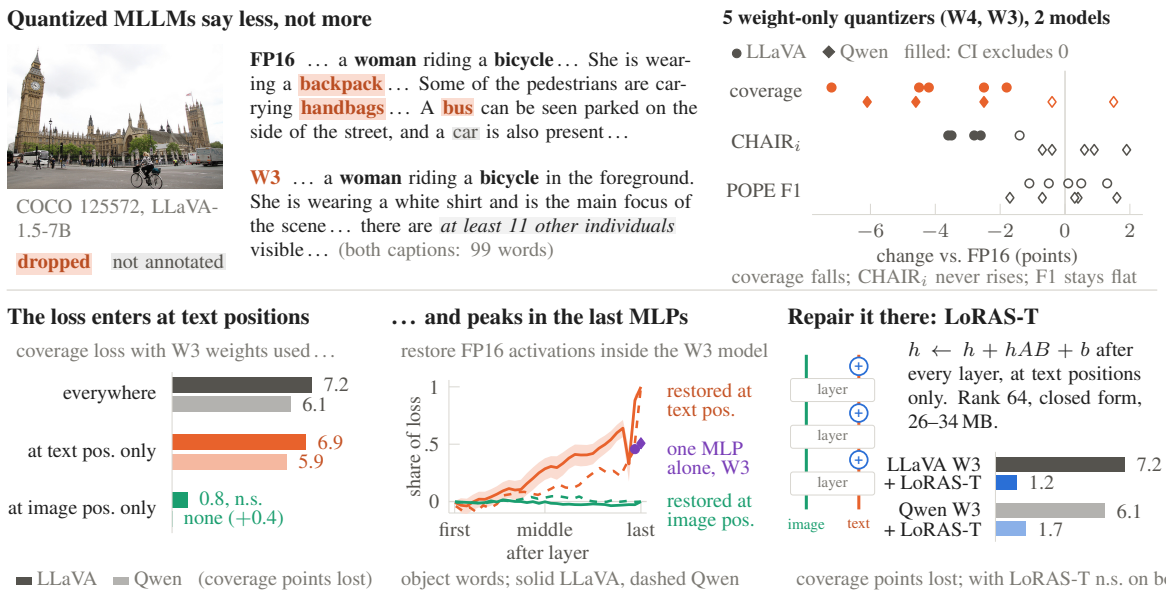


Figure 1: Overview. **Top:** with 3-bit weights, an MLLM keeps its caption length but names fewer of the objects present and invents none; across quantizers and both models, coverage falls while CHAIR$_i$ and POPE F1 do not move. **Bottom:** the loss enters at text positions (§5), builds up to the last MLPs (§5.1), and is removed by a correction at text positions only (§6).

duce additional hallucination (Kundu et al., 2025). Together, these suggest a *lexical fallback* account of quantized MLLMs: quantization noise weakens the visual tokens, the decoder stops relying on them, and the model falls back on its language priors, inventing plausible but absent objects. If it held, quantized MLLMs would need hallucination defences, and quantizers would need to protect the image tokens.

We test this account directly, at 3 bits, where low-bit quantization is heading and where its damage becomes measurable, with measures that separate invented objects from missed ones and causal interventions that isolate where the damage enters. It is wrong in both its direction and its location.

Contributions. On LLaVA-1.5-7B and Qwen2.5-VL-7B, two MLLMs with different vision encoders, decoders and tokenization, we show:

1. **Quantized MLLMs say less, not more.** With 3-bit weights, captions keep their length but name up to 7.2 (LLaVA) and 6.1 (Qwen) points fewer of the objects present, small objects first. No weight-only quantizer significantly raises hallucination on either model, and POPE, the standard benchmark, registers none of the loss (Figure 1).

2. **The loss enters at text positions.** Using 3-bit weights only while the decoder processes text tokens reproduces the full loss; using them only at image tokens causes none. This establishes by controlled intervention what prior work inferred from gradients and activation statistics.

3. **It builds up across depth and peaks in the last MLPs.** Restoring full-precision activations and weights inside the quantized decoder, we show that image positions carry almost none of the loss, that it accumulates layer by layer at text positions, and that it sits in MLP rather than attention weights: the last MLPs alone reproduce about half of it. This is the first localization of quantization damage by layer, component and token position in an MLLM.

4. **Repairing it where it is.** LoRAS-T, a low-rank correction of the decoder's hidden states at text positions, fitted in closed form without gradients, removes the significant coverage loss of 3-bit round-to-nearest on both models. Applied at every position instead, with the same parameters and data, the same correction recovers less coverage (LLaVA) or breaks yes/no answering (Qwen): where the repair is placed matters.

These findings change how compressed MLLMs should be evaluated and built. Hallucination met-

	MLLM Grounding	Omission vs. hall.	Shift found		Image vs. text	Layer × comp.	Matched noise	Repair acts on
PTQ methods for MLLMs								
Q-VLM	✓	✗	✗	–	✗	✗	✗	–
MBQ, VLMQ, QIG	✓	✗	✗	–	◐	✗	✗	–
MQuant, SplitQ	✓	✗	✗	–	◐	✗	✗	–
LUQ	✓	◐	✗	–	◐	✗	✗	–
Balanced Fitting	✓	✗	✗	–	◐	◐	✗	–
Evaluations of quantized MLLMs								
Das et al.	✓	✗	✗	–	✗	◐	✗	–
LVLM-Compress	✓	◐	✗	no rise[a]	✗	✗	✗	–
GHOST-Q	✓	✓	✗	both ways	✗	✗	✗	–
Saha et al.	✓	✓	✓	none	✗	◐	✗	–
Text-only LLMs								
Gong et al.	✗	–	–	–	–	✗	✓	–
Chang et al.	✗	–	–	–	–	✓	✗	–
Hu & Ramachandran	✗	–	–	–	–	◐	✗	–
QwT	✗	–	–	–	–	✗	✗	all tokens
Low-rank comp.	✗	–	–	–	–	✗	✗	weights
Ours	✓	✓	✓	**omission**	✓	✓	✓	**text pos.**

Table 1: Prior work vs. ours. **Grounding**: hallucination measured in captions and yes/no questions (✓) or yes/no only (◐). **Image vs. text**: damage attributed to a token type by quantizing only that type (✓) or via gradients, statistics or loss (◐). **Layer × comp.**: full precision restored per layer and component (✓) or coarser swaps (◐). **Matched noise**: equal-magnitude perturbation control. –: not applicable. [a]KV-cache compression.

rics reward a model that says less, so grounding evaluations must report how many present objects are named. And quantizers and repairs for MLLMs should target the decoder's computation at text positions, not protect the visual tokens.

2 Related Work

Table 1 positions our work along the dimensions of our contributions.

Quantizing MLLMs. PTQ methods for MLLMs search block-wise quantizers (Wang et al., 2024), reweight calibration by modality or token saliency (Li et al., 2025; Xue et al., 2025; Xiang et al., 2026), separate modality-specific activation ranges and outliers (Yu et al., 2025; Zhong et al., 2026), allocate bits by layer entropy (Bhatnagar et al., 2026), or balance per-layer fitting (Kang et al., 2026). Several infer that text tokens are more sensitive than image tokens, from gradients (Li et al., 2025; Xue et al., 2025) or activation statistics (Zhong et al., 2026). None tests this by intervention on model outputs, and all report benchmark accuracy rather than grounding; LUQ reports POPE F1 only. They build on LLM quantizers: GPTQ (Frantar et al., 2023), AWQ (Lin et al., 2024), SmoothQuant (Xiao et al., 2023), rotation (Ashkboos et al., 2024) and the NF4 format (Dettmers et al., 2023).

Evaluating quantized MLLMs. Das et al. (2026) quantize the vision encoder and language model separately and find the language model most sensitive. LVLM-Compress-Bench finds no major rise in hallucination under KV-cache compression (Kundu et al., 2025); GHOST-Q finds item-level POPE and AMBER answers flipping in both directions under unchanged scores (Rehman and Shafique, 2026); Saha et al. (2026) find yes/no margins shifting at 4 bits before answers flip, with no change in CHAIR or object recall. We are the first to find a directional shift in grounding under quantization, toward omission, and to show that standard metrics hide it.

Mechanisms of quantization damage. In text-only LLMs, perturbations of matched magnitude show that the size of quantization error matters more than its distribution (Gong et al., 2024); patching full-precision activations into 3-bit models places most of the damage in MLP outputs (Chang et al., 2025); and raising single layers to higher precision finds the recoverable damage spread across layers (Hu and Ramachandran, 2026). Activation patching (Meng et al., 2022; Zhang and Nanda, 2024) has located factual recall in full-precision MLLMs (Basu et al., 2024). In quantized MLLMs, localization has been limited to raising blocks from 4 to 8 bits (Saha et al., 2026) or loss-level restoration (Kang et al., 2026). We restore full precision per layer, component and token position in quantized MLLMs, and bring the matched-noise control to grounding.

Error compensation and steering. Quantization error is compensated by low-rank terms on the weights (Zhang et al., 2024, 2025a; Liu et al., 2024c; Scetbon and Hensman, 2024) or, closest to us, by a closed-form linear correction of every token's block output (Fu et al., 2025). Activation steering edits hidden states to change behaviour (Li et al., 2023a; Turner et al., 2023; Wu et al., 2024), including against MLLM hallucination (Liu et al., 2025a). LoRAS-T is the first training-free correction restricted to the token positions where

quantization damage enters.

Object hallucination. POPE (Li et al., 2023b) asks yes/no questions about objects, CHAIR (Rohrbach et al., 2018) counts invented objects in captions, and AMBER (Wang et al., 2023) adds coverage. Hallucination falls when captions say less (Yue et al., 2024), which is why CHAIR alone cannot detect omission. Full-precision MLLMs already miss small details (Zhang et al., 2025b); we show that quantization removes them first.

3 Setup

Models. LLaVA-1.5-7B (Liu et al., 2024a) encodes an image with CLIP ViT-L/14 into 576 *image tokens*, which a 32-layer Vicuna decoder reads together with the *text tokens* of the prompt and of its own answer. Qwen2.5-VL-7B (Bai et al., 2025) uses a dynamic-resolution ViT (about 390 image tokens per COCO image) and a 28-layer decoder with grouped-query attention and multimodal rotary positions. We quantize the decoder only (6.5B parameters in each model), which is where both models hold nearly all their weights.

Measuring grounding. In *captioning* ("Please describe this image in detail."), let G_c be the objects annotated in image c and M_c the objects its caption names, among the 80 COCO categories. Two rates separate the two failure modes:

$$\text{CHAIR}_i = \frac{\sum_c |M_c \setminus G_c|}{\sum_c |M_c|}, \qquad \text{Cov} = \frac{\sum_c |G_c \cap M_c|}{\sum_c |G_c|}.$$

CHAIR_i (Rohrbach et al., 2018) is the share of named objects that are absent (hallucination); *coverage* is the share of present objects that are named (grounding recall). A model that says less lowers both, so they must be read together. We also track which of the objects FP16's caption names disappear from the quantized caption, by object size. In *POPE* (Li et al., 2023b), the model answers "Is there a [object] in the image?" for present and absent objects; we report F1, the share of "yes" answers, and the AUROC of $P(\text{yes})$, which is independent of the yes/no threshold. WikiText-2 perplexity (Merity et al., 2017) measures the language model's health on its own.

Hypotheses. Lexical fallback (**H1**) predicts rising CHAIR_i and yes-ratio with stable coverage. Omission (**H2**) predicts falling coverage without rising CHAIR_i. If quantization merely damages the language model (**H0**), grounding should degrade in step with perplexity.

Quantizers. Wb denotes b-bit weights with 16-bit activations. We compare NF4, round-to-nearest (RTN), GPTQ, AWQ, and rotation with GPTQ weights at W4A4, re-implemented on one simulated-quantization substrate so that methods differ only in their algorithm (groups of 128 weights), and calibrated on WikiText-2 or on COCO images with captions. Our re-implementations match or exceed the published gains over RTN in most settings, and simulated NF4 matches the real bitsandbytes kernels (Appendix A). We focus on 3-bit weights, the regime the field is moving into and the first in which damage becomes measurable; NF4 and 4-bit GPTQ serve as near-lossless references, and W4A4 shows the separate effect of quantizing activations.

Protocol. Captions (greedy, at most 128 tokens) are scored on 300 COCO val2014 images and POPE on 1,800 questions over 100 images; all corrections are calibrated on a disjoint image pool. Every comparison is paired, quantized minus FP16 on identical items, and tested with an image-level bootstrap (Efron and Tibshirani, 1994); arrows in the tables mark changes whose 95% interval excludes zero.

4 Quantized MLLMs Say Less, Not More

Table 2 compares seven deployed quantization settings with FP16 on both models.

Coverage falls; hallucination does not rise. Every 3-bit quantizer lowers coverage on LLaVA, by 2.5 to 7.2 points, and three of four do on Qwen, by up to 6.1. Captions keep their length (FP16: 93 and 107 words) but name fewer of the objects that are there. Meanwhile CHAIR_i rises significantly for no weight-only quantizer on either model; on LLaVA it *falls* in four of seven settings, because a caption that names fewer objects also names fewer wrong ones. Figure 1 shows a typical case: two 99-word captions of the same street scene, of which the 3-bit one drops the backpack, the handbags and the bus and replaces them with "at least 11 other individuals". The pattern is not specific to these settings: across a wider sweep of 16 quantization settings on LLaVA, coverage is below FP16 in all 16 (Appendix B). Across both models, every setting moves toward fewer objects named, none toward more objects invented (Figure 2a): the data match omission (H2) and contradict lexical fallback (H1).

Change vs. FP16 (points)	LLaVA-1.5-7B Cov	LLaVA-1.5-7B $CHAIR_i$	LLaVA-1.5-7B F1	LLaVA-1.5-7B small	LLaVA-1.5-7B PPL	Qwen2.5-VL-7B Cov	Qwen2.5-VL-7B $CHAIR_i$	Qwen2.5-VL-7B F1	Qwen2.5-VL-7B small	Qwen2.5-VL-7B PPL
FP16 (absolute)	*74.4*	*22.0*	*85.4*		*7.27*	*55.9*	*10.9*	*83.3*		*8.28*
NF4 (bitsandbytes)	−0.5	+1.6	+0.2	11	1.02×	+0.3	−0.6	**+1.8**↑	19	1.01×
W4 GPTQ	**−1.8**↓	**−2.6**↓	+0.1	16	1.02×	+1.5	−0.7	−0.7	16	1.02×
W4A4 rotation+GPTQ	−0.2	−0.1	+0.2	14	–	−1.9	−1.3	**−6.3**↓	26	1.21×
W3 GPTQ (image+caption)	**−2.5**↓	**−3.6**↓	−0.5	20	1.36×	−0.4	+1.9	+1.6	17	1.36×
W3 AWQ	**−4.2**↓	−1.4	−1.1	22	1.17×	**−2.5**↓	+0.9	+0.4	25	1.24×
W3 GPTQ	**−4.5**↓	**−2.8**↓	+0.5	24	1.17×	**−4.6**↓	+0.6	−1.7	35	1.13×
W3 RTN	**−7.2**↓	**−3.5**↓	+1.3	28	1.26×	**−6.1**↓	−0.4	+0.3	36	1.29×
Matched noise: Gaussian weight noise with RTN's error energy										
W4 noise	**−2.7**↓	**−3.0**↓	−0.3	13	–	−0.9	+1.0	−0.7	18	1.05×
W3 noise	**−9.7**↓	**−4.6**↓	−1.1	31	–	**−4.2**↓	+0.4	**−3.6**↓	25	1.29×
W3 RTN weights at one position type, FP16 weights at the other (§5)										
image positions only	−0.8	−0.6	+0.4	13		+0.4	+0.5	−1.2	12	
text positions only	**−6.9**↓	**−4.0**↓	+0.8	28		**−5.9**↓	−0.6	−1.6	33	

Table 2: Grounding under quantization, paired against FP16 (300 captions; 1,800 POPE questions). Arrows: 95% interval excludes 0; grey: not significant. **small**: % of the small objects (<1% of the image) named by FP16 that the quantized caption drops. **PPL**: WikiText-2 perplexity relative to FP16. Coverage falls; $CHAIR_i$ never rises.

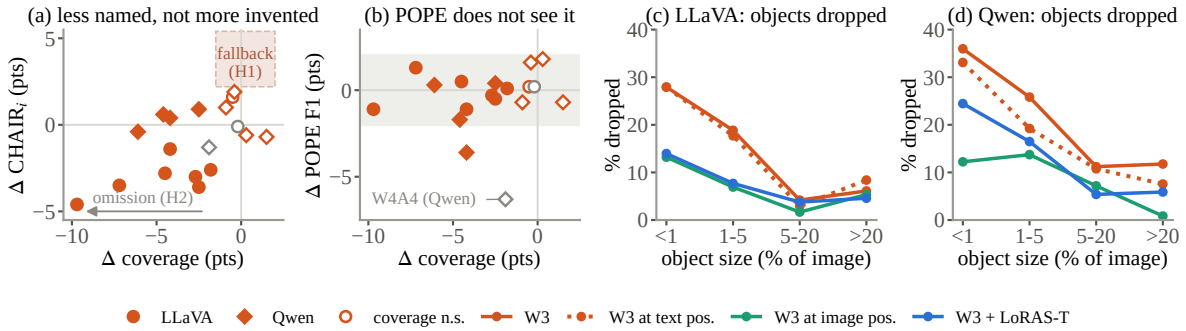


Figure 2: The omission signature, on the settings of Table 2. **(a)** Quantization moves models left (fewer present objects named), not up into the region lexical fallback predicts. **(b)** POPE F1 stays within ±2 points (grey band) except when activations are quantized. **(c, d)** Share of FP16-named objects that are dropped, by object size: 3-bit weights drop small objects first; using them only at text positions reproduces this, only at image positions does not, and LoRAS-T removes most of it.

POPE does not see it. POPE F1 stays within 2 points of FP16 for every weight-only quantizer on both models. Underneath, the yes-ratio does move, in both directions (LLaVA: +6.1 for W3 AWQ, −3.1 for W3 RTN), with precision and recall shifting in opposite directions and AUROC nearly unchanged (at most 1.1 points): quantization moves the yes/no threshold, not the ranking of present over absent objects. The only large POPE drop comes from quantizing activations, W4A4 on Qwen (−6.3; Figure 2b). A benchmark that asks about one object at a time does not register a model that has stopped mentioning objects on its own.

Small objects are lost first. At 3 bits, 28% (LLaVA) and 36% (Qwen) of the small objects that FP16 names disappear from the quantized captions, against 6% and 12% of the large ones (>20% of the image). The drop rate falls steadily with object size on both models (Figure 2c, d): quantization first removes the details the model was least sure of.

This is not language-model damage. If omission were a by-product of a damaged language model (H0), the settings with the highest perplexity would lose the most coverage. They do not. On both models, the 3-bit quantizer that damages the language model most, GPTQ calibrated on images and captions (1.36× perplexity), loses the *least* coverage (−2.5 and −0.4), and on Qwen, AWQ has higher perplexity than GPTQ (1.24× vs. 1.13×) but loses half as much coverage. Grounding loss is not a proxy for perplexity.

POPE F1 change (RTN weights)	LLaVA	Qwen
W4A8	+0.9	−0.1
W4A6	+0.8	**−11.6**↓
W4A5	**−9.3**↓	broken
W4A4	**−58.8**↓	broken
A4 at image positions only	−1.1	**−30.0**↓
A4 at text positions only	**−73.6**↓	broken

Table 3: 4-bit activations break both models at text positions; at image positions they leave LLaVA intact and Qwen degraded but fluent. Other positions at 8 bits; broken: >95% of answers unparseable; LLaVA on 60 images.

It is a response to weight error, not to rounding. Replacing RTN's rounding error by Gaussian noise of exactly the same energy in every weight group also produces omission (-9.7 and -4.2 at 3 bits), but no consistent change in CHAIR_i. Losing objects is the decoder's generic response to weight error of this size; inventing them is not. Quantizers differ in how much error they leave, and GPTQ, which leaves less, also loses less coverage than RTN; but the direction of the damage is set by the decoder, not by the quantizer.

5 Where the Loss Enters

Lexical fallback places the damage in the visual tokens. We test where it enters with two causal interventions that differ in granularity: one switches quantization on and off by token position, the other restores full precision inside the network one site at a time.

Image or text positions? Every linear layer of the decoder processes image and text tokens alike, so we can quantize it for one position type and not the other: outputs at image positions are computed with 3-bit weights and outputs at text positions with FP16 weights, or vice versa. The result is unambiguous on both models (Table 2, bottom). With 3-bit weights only at image positions, coverage stays at FP16 level (-0.8, $+0.4$) and small-object drops stay at the level of the mildest settings. With 3-bit weights only at text positions, the full loss appears: -6.9 against -7.2 for the fully quantized LLaVA, -5.9 against -6.1 for Qwen. Activations tell the same story (Table 3): 4-bit activations at text positions break both models completely, while at image positions they leave LLaVA intact and Qwen fluent. Text tokens carry the largest activation outliers in both models (up to $|x| = 1564$ in LLaVA and 2288 in Qwen, against 310 at Qwen's image positions), consistent with massive activations (Sun et al., 2024) and with gradient evidence that language tokens are the more sensitive (Li et al., 2025). The visual tokens survive quantization; what fails is the computation at text positions that turns them into words. Reduced attention to the image is not the mechanism either: 4-bit weights lower LLaVA's image attention by 14% but *raise* Qwen's by 14%, while both models lose objects (Appendix C).

5.1 Inside the decoder: cross-precision patching

Position-selective quantization says *which tokens* carry the loss, not *where in the decoder* it arises. We locate it by causal patching (Meng et al., 2022; Zhang and Nanda, 2024) between the FP16 model and its quantized copy. For 100 images, we teacher-force FP16's own caption through both models and measure, for each caption token, the log-probability lost by quantization. An intervention p that restores part of the network to full precision recovers a share $R(p)$ of that loss:

$$R(p) = \frac{\sum_t (\ell_t^p - \ell_t^q)}{\sum_t (\ell_t^{\text{fp}} - \ell_t^q)},$$

where ℓ_t^{fp}, ℓ_t^q and ℓ_t^p are token t's log-probabilities under FP16, the quantized model and the patched model. We restore activations (the residual stream, attention output or MLP output of one layer, at all image or all text positions) and weights (one layer's attention or MLP, either restored in the quantized model or quantized alone in the FP16 model). We separate object words, the first sub-token of each annotated object the caption names, from all other words.

Image positions carry almost none of the loss. Restoring every image position to FP16, at any layer and site, recovers at most 4% (LLaVA) and 5% (Qwen) of it (Figure 3, green). The image tokens are computed with quantized weights in this model, yet their representations are not what the text positions fail to read: the visual information arrives intact at every depth.

The loss builds up across depth and peaks in the last MLPs. Restoring the text-position residual stream after layer ℓ recovers a share that grows steadily with ℓ (orange): the loss accumulates through the decoder rather than entering at one point, and a large part arises in the final layers. Quantizing one layer at a time places it in MLP

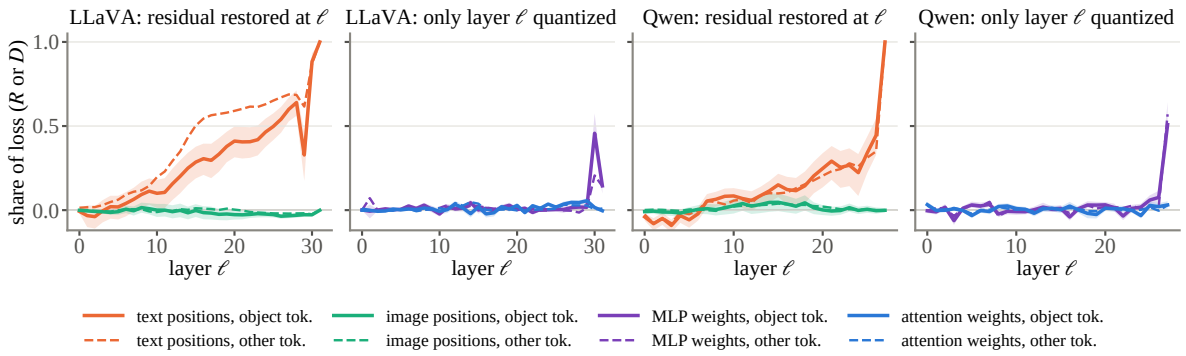


Figure 3: Cross-precision patching on W3 RTN (100 FP16 captions per model). *Residual restored at ℓ*: share of the log-probability loss recovered when the residual stream after layer ℓ is set to FP16 at all text (orange) or image (green) positions, all later layers quantized. *Only layer ℓ quantized*: share of the loss reproduced by quantizing one layer's MLP (purple) or attention (blue) in an FP16 model. Solid: object words; dashed: other words.

weights rather than attention: summed over layers, single MLPs reproduce 0.86 (LLaVA) and 0.74 (Qwen) of the object-word loss, single attention blocks 0.35 and 0.11. One site dominates: quantizing only the last MLP of Qwen, 3% of the decoder's weights, reproduces 51% of the object-word loss, and only the penultimate MLP of LLaVA, 46% (purple). These results extend to MLLMs what Chang et al. (2025) found for perplexity in text-only LLMs, and add what is specific to them: the damage sits at text positions.

Omission emerges in generation. Per token, quantization costs object words about as much log-probability as other words (0.13 vs. 0.09 nats on LLaVA, 0.22 vs. 0.27 on Qwen) and flips FP16's top choice equally often for both (14–18% of tokens). Quantization does not target the names of objects; it perturbs every token at text positions, and in free generation these perturbations compound into captions that name fewer objects. This matches the matched-noise result (§4), and it tells us what a repair must do: correct the text-position computation as a whole, across depth. Saha et al. (2026) find earlier blocks most important for yes/no margins at 4 bits; for the objects a model names in its captions, our full-precision restoration points to the end of the decoder.

6 Repairing the Loss at Text Positions

LoRAS-T. The loss enters at text positions and accumulates across layers, so we correct the hidden state of every text token, prompt and generated alike, after every decoder layer, and leave image tokens untouched. LoRAS-T (Low-Rank Activation Steering at Text positions) adds to the hidden state $h_\ell \in \mathbb{R}^d$ of layer ℓ a rank-64 linear correction,

$$h_\ell \leftarrow h_\ell + h_\ell A_\ell B_\ell + b_\ell,$$

with $A_\ell \in \mathbb{R}^{d\times 64}$ and $B_\ell \in \mathbb{R}^{64\times d}$, fitted to map the quantized model's hidden states onto FP16's on the same inputs, in closed form by reduced-rank ridge regression (Scetbon and Hensman, 2024; Fu et al., 2025). With centred quantized states X and residuals $R = H^{\text{FP16}} - H^{Q}$, we compute $M = (X^\top X + \lambda I)^{-1} X^\top R$, take the top-64 eigenvectors V of $M^\top X^\top X M$, and set $A = MV$, $B = V^\top$. Layers are fitted in order, each with all earlier corrections active, so that every correction sees the inputs it will see at inference. Calibration uses 192 images, each with FP16's own caption and two yes/no object questions answered by FP16. Fitting takes 30 minutes on one GPU with no gradient steps, and LoRAS-T adds 34 MB (LLaVA) or 26 MB (Qwen), under 0.3% of the decoder.

LoRAS-T removes the loss on both models. On 3-bit RTN, LoRAS-T recovers 83% (LLaVA) and 72% (Qwen) of the coverage loss, leaving no significant difference from FP16 on either model (Table 4), and it halves LLaVA's small-object drops, from 28% to 14%. CHAIR_i returns to the FP16 level, not above it: the correction restores FP16's balance between naming and inventing objects rather than buying coverage with invented ones. POPE is unaffected (F1, yes-ratio and AUROC all within 1 point of FP16 on both models). On LLaVA, 3-bit RTN with LoRAS-T differs significantly from FP16 on no measure, while 4-bit GPTQ, which stores 0.8 GB more, still loses cover-

vs. FP16	Cov	CHAIR$_i$	F1	small	rec.
LLaVA-1.5-7B					
W3 RTN	**−7.2**↓	**−3.5**↓	+1.3	28	
+ LoRAS-T	−1.2	+0.9	+0.5	**14**	**83%**
+ all positions	**−2.9**↓	−0.2	+0.5	18	61%
W3 GPTQ	**−4.5**↓	**−2.8**↓	+0.5	24	
+ LoRAS-T	**−2.5**↓	+0.8	−0.7	15	44%
W4 GPTQ (+0.8 GB)	**−1.8**↓	**−2.6**↓	+0.1	16	
Qwen2.5-VL-7B					
W3 RTN	**−6.1**↓	−0.4	+0.3	36	
+ LoRAS-T	−1.7	+0.2	−0.3	**24**	**72%**
+ all positions	−1.9	−0.8	**−3.4**↓	25	69%

Table 4: LoRAS-T on the evaluation set of Table 2. **rec.**: share of the coverage loss recovered, paired against the uncorrected model. "All positions": the same corrector (rank, layers, data) fitted and applied at image positions too.

age; on 3-bit GPTQ, LoRAS-T recovers a further 44% of the loss.

Placement, not capacity. Is it the text-side placement that helps, or only the added parameters? We fit the same corrector, with the same rank, layers and calibration data, at every position, image tokens included. On neither model does this help, and on both it costs. On LLaVA it recovers less coverage, 61% instead of 83%, leaving a significant 2.9-point loss and more small-object drops (18% vs. 14%). On Qwen it recovers as much coverage but creates a loss of its own: POPE recall falls by 5.0 points and F1 by 3.4 as the corrected model starts denying objects that are present, which text-only LoRAS-T does not do. Image positions carry almost none of the quantization loss (§5.1); correcting them spends capacity where nothing is broken and disturbs what was intact.

The correction must see questions. Calibrated on captions alone, LoRAS-T still recovers coverage, but it shifts POPE's yes-ratio (by 8.2 points on W3 GPTQ for LLaVA) while AUROC stays at the uncorrected level: having never seen a question prompt, the correction moves the threshold at which the model answers "yes". Adding yes/no questions answered by FP16 to the calibration removes the shift, bringing the yes-ratio within 0.6 points of FP16.

7 Conclusion

Quantization does not make MLLMs hallucinate; it makes them see less. On LLaVA-1.5 and Qwen2.5-VL, standard quantizers remove present objects from captions, small ones first, while hallucination metrics hold steady or improve and POPE stays flat. The visual tokens survive; the loss enters at text positions, builds up across the decoder and peaks in its last MLPs. A closed-form correction placed there, and only there, brings 3-bit models back to full-precision grounding. Evaluations of compressed MLLMs should measure what a model leaves out, and quantizers should protect the text side of the decoder.

Limitations

Our quantizers run on a common simulated substrate rather than real low-bit kernels (except NF4), and the KV cache is not quantized. We study two 7B MLLMs on COCO with greedy decoding; larger models, other domains and other decoding strategies may behave differently. Our grounding measures rely on COCO annotations, which miss unannotated objects. Patching scores teacher-forced FP16 captions and single sites, not free generation or interactions between sites. LoRAS-T is distilled from FP16 and inherits its errors. Future work should test real kernels and larger models, and turn the localization into quantizers that spend precision on the text side and the last MLPs.

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A Validation of the Quantizers

GPTQ and AWQ, calibrated on WikiText-2, close 35–56% of RTN's 3-bit perplexity gap, matching or exceeding the published share, except AWQ on Qwen; rotation reduces W4A4 perplexity from collapse to near W3 levels on both models (Table 5). Simulated NF4 gives the same POPE metrics as the real bitsandbytes kernels and coverage within 0.1 points (Table 2).

	LLaVA		Qwen	
WikiText-2	PPL	closed	PPL	closed
FP16	7.27		8.28	
NF4 (simulated)	7.39		8.38	
W4 RTN	7.50		8.83	
W4 GPTQ	7.45	22%	8.43	72%
W3 RTN	9.14		10.66	
W3 AWQ	8.47	36%	10.29	15%
W3 GPTQ	8.49	35%	9.32	56%
W4A4 RTN	1001		59989	
W4A4 rotation	10.6		11.2	

Table 5: Perplexity of the quantized decoders. **Closed**: share of RTN's gap to FP16 removed by the method; for Llama-2-7B, AWQ reports 35% at 3 bits (Lin et al., 2024).

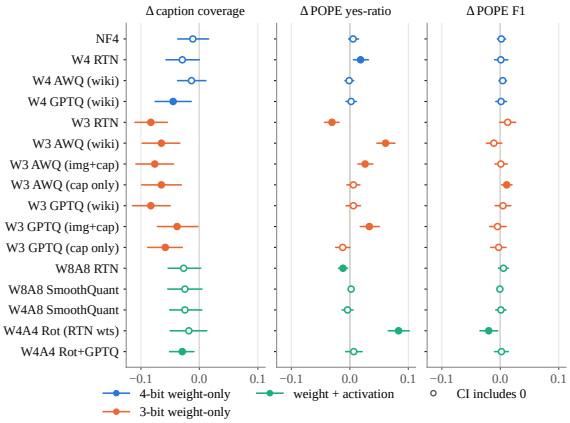


Figure 4: LLaVA-1.5, 16 settings, paired against FP16 (100 captions, 1,800 POPE questions); filled: 95% interval excludes 0. Coverage falls in all 16 settings while F1 stays within ±0.02.

B A Wider Sweep of Quantization Settings

Figure 4 adds RTN and AWQ at 4 bits, AWQ and GPTQ at 3 bits with each calibration source, W8A8 with and without SmoothQuant (Xiao et al., 2023), W4A8 SmoothQuant and W4A4 rotation with RTN weights. Coverage is below FP16 in every setting (sign test, $p = 3 \cdot 10^{-5}$); CHAIR_i rises significantly only for W4A4 rotation with RTN weights, the setting with the most damaged language model. On Qwen, 3-bit RTN in the same sweep loses 9.0 coverage points.

C Attention to the Image

4-bit weights lower LLaVA's attention to the image but raise Qwen's (Table 6), while both models lose objects. Reduced visual attention is a model-specific correlate of quantization, not the mechanism of omission.

Image attention vs. FP16	POPE	captions
LLaVA, W4 GPTQ	**−14.0%**↓	**−18.3%**↓
Qwen, W4 GPTQ	**+13.9%**↑	**+8.8%**↑

Table 6: Share of generated tokens' attention on image tokens, averaged over layers and heads (100 images per task).